

Lab 8 : Challenge

Challenge 2

`nmap -F 192.168.44.131`

-F: Enables fast scan mode, restricting the sweep to the top 100 most common ports rather than the default port range.

192.168.44.131: Specifies the target IPv4 address for the scan.

```
(kali㉿kali)-[~]
$ nmap -F 192.168.44.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 04:07 -0400
Nmap scan report for 192.168.44.131
Host is up (0.00029s latency).
Not shown: 82 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
8009/tcp  open  ajp13
MAC Address: 00:0C:29:C1:4C:6C (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds
```

Identified 18 open TCP ports, including key network services such as FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), NetBIOS/SMB (139/445), NFS (2049), MySQL (3306), PostgreSQL (5432), VNC (5900), and X11 (6000).

The resolved MAC address indicates the target host is running within a VMware virtualized environment.

Port enumeration immediately isolates immediate exposure vectors running legacy or misconfigured services. Notably, subsequent version identification confirms the presence of vsftpd 2.3.4, a service containing a critical, known vulnerability.

Challenge 5

Triggers ICMP Echo Request packets to verify host availability and calculate round-trip network latency.

ping 192.168.44.131

```
(kali㉿kali)-[~]  
$ ping 192.168.44.131  
PING 192.168.44.131 (192.168.44.131) 56(84) bytes of data.  
64 bytes from 192.168.44.131: icmp_seq=1 ttl=64 time=0.528 ms  
64 bytes from 192.168.44.131: icmp_seq=2 ttl=64 time=0.381 ms  
64 bytes from 192.168.44.131: icmp_seq=3 ttl=64 time=0.346 ms  
64 bytes from 192.168.44.131: icmp_seq=4 ttl=64 time=0.437 ms  
64 bytes from 192.168.44.131: icmp_seq=5 ttl=64 time=0.499 ms  
^C  
— 192.168.44.131 ping statistics —  
5 packets transmitted, 5 received, 0% packet loss, time 4081ms  
rtt min/avg/max/mdev = 0.346/0.438/0.528/0.068 ms
```

0% Packet Loss: Confirms host availability and connection stability.

RTT (~0.4 ms): Indicates minimal round-trip time latency, consistent with local area network (LAN) or virtualized environments.

TTL (64): Aligns with the default initial hop limit standard for Linux/Unix-based operating systems.

Verifies operational host status prior to deep-scope scanning. While firewalls frequently drop ICMP traffic to obscure host presence, this responsive baseline establishes an active target profile.

Challenge 9

Establishes a raw TCP connection to Port 21 (FTP) to retrieve and log the service's welcome banner.

nc 192.168.44.131 21

```
(kali㉿kali)-[~]  
$ nc 192.168.44.131 21  
220 (vsFTPd 2.3.4)
```

Validates the precise version of the active FTP server.

The vsftpd 2.3.4 daemon contains a notorious backdoor vulnerability. Appending a specific string—specifically a smiley face emoticon :)—to the authentication username triggers a malicious listener, opening a root shell on Port 6200.

Challenge 16

-sV: Enables service version detection, interrogating identified open ports to determine exact daemon names, product variants, and version signatures.

`nmap -sV 192.168.44.131`

```
[kali@kali]~$ nmap -sV 192.168.44.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 03:59 -0400
Stats: 0:00:07 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 47.83% done; ETC: 03:59 (0:00:07 remaining)
Nmap scan report for 192.168.44.131
Host is up (0.00096s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11 (Xes App) (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:C1:4C:6C (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.28 seconds
```

Why pros avoid NAT for pentesting

Because NAT:

- Hides targets
- Blocks inbound probes
- Breaks scripts like `smb-enum-users`
- Doesn't simulate real networks

Bottom line

- ❌ NAT cannot be "unblocked" for full scans
- ⚠️ Port forwarding = partial hack, not ideal
- ✅ Host-only / Bridged = proper pentesting

If you want, I can design your full home lab network for you. DM me on Twitter.

scans work perfectly—like a real red team.

Challenge 17

-O: Enables operating system (OS) detection, transmitting a series of specialized TCP/IP payloads to analyze target network stack behaviors against a known database of OS signatures.

`nmap -O 192.168.44.131`

Identifies the target kernel signature within the Linux 2.6.9 – 2.6.33 range.

This legacy kernel architecture dates from approximately 2004 to 2010. For reference, vulnerable test environments like Metasploitable 2 utilize kernel version 2.6.24.

Outdated kernel versions expose a significant attack surface susceptible to local privilege escalation (LPE) exploits. Furthermore, confirming a non-Windows architecture narrows the exploit scope, steering payload selection exclusively toward Linux-compatible vectors.

```

(kali㉿kali)-[~]
$ nmap -O 192.168.44.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-04 04:01 -0400
Nmap scan report for 192.168.44.131
Host is up (0.00071s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:C1:4C:6C (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 2.09 seconds

```

Challenge 19

Queries the portmapper service (Port 111) to list all registered RPC services and corresponding ports.

`rpcinfo -p 192.168.44.131`

```
—(kali@kali)-[~]  
—$ rpcinfo -p 192.168.44.131  
program vers proto  port  service  
100000    2    tcp    111   portmapper  
100000    2    udp    111   portmapper  
100024    1    udp    40173 status  
100024    1    tcp    35789 status  
100003    2    udp    2049  nfs  
100003    3    udp    2049  nfs  
... 100003    4    udp    2049  nfs  
100021    1    udp    36796 nlockmgr  
100021    3    udp    36796 nlockmgr  
100021    4    udp    36796 nlockmgr  
100003    2    tcp    2049  nfs  
100003    3    tcp    2049  nfs  
100003    4    tcp    2049  nfs  
100021    1    tcp    51062 nlockmgr  
100021    3    tcp    51062 nlockmgr  
100021    4    tcp    51062 nlockmgr  
100005    1    udp    35635 mountd  
100005    1    tcp    41293 mountd  
100005    2    udp    35635 mountd  
100005    2    tcp    41293 mountd  
100005    3    udp    35635 mountd  
100005    3    tcp    41293 mountd
```